Design and Optimization of a Low-Power Low-Noise CMOS ECG Front-End IC with Enhanced Gain Using Transistor Sizing and Body Modeling Circuit

Jairus S. Solomon
School of Electronics Engineering
Vellore Institute of Technology
Vellore, India
jairus.samraj.solomon@gmail.com

Subham P. Gupta
School of Electronics Engineering
Vellore Institute of Technology
Vellore, India
shubham01dec04@gmail.com

Vinayak S. Salunke
School of Electronics Engineering
Vellore Institute of Technology
Vellore, India
salunkevinayak2005@gmail.com

Dr. Md. Sadullah
School of Electronics Engineering
Vellore Institute of Technology
Vellore, India

***Abstract*—Electrocardiogram (ECG) signal acquisition requires amplification of extremely low-amplitude bio-signals in the presence of significant noise and interference. This paper presents the design and optimization of a low-power, low-noise CMOS analog front-end (AFE) for ECG applications. The proposed system consists of an instrumentation amplifier, a two-stage operational transconductance amplifier (OTA), a Driven Right Leg (DRL) circuit, and filtering stages for effective ECG signal conditioning..**

The novelty of this work lies in optimizing transistor sizing to improve overall gain and noise performance, along with the implementation of a realistic body impedance model to simulate realistic physiological signal conditions. These improvements enhance both system performance and the accuracy of simulation results.

The circuit is designed and simulated using Cadence Virtuoso and Spectre simulator. The proposed design achieves high gain, excellent common-mode rejection ratio (CMRR), and stable frequency response while maintaining low power consumption, making it suitable for wearable biomedical applications.

***Index Terms*—ECG, CMOS, Low Power, Low Noise, OTA, CMRR, DRL Circuit, Transistor Sizing, Analog Front-End**

I. Introduction

Electrocardiogram (ECG) signal is very important in modern healthcare systems. It is used to monitor the electrical activity of the heart and helps in detecting heart related problems like arrhythmia and heart failure. Nowadays many wearable devices are used for continuous health monitoring so ECG signal acquisition becomes more important. But the main problem is that ECG signals are very weak and usually in the range of microvolts to millivolts and they are easily affected by different types of noise like power line interference motion artifacts and thermal noise [1], [2].

To handle this problem an analog front end circuit is required. This circuit amplifies the weak ECG signal and also removes noise. Designing this type of circuit is not easy because we need high gain low noise low power consumption and also high common mode rejection ratio. These requirements are very important especially for wearable devices where battery life is limited [3], [4].

Many researchers have worked on improving ECG front end circuits. Some common methods include using capacitive feedback and pseudo resistors to control gain and frequency. Inverter based OTA is also used because it gives better efficiency at low power. Another important part is the Driven Right Leg circuit which helps in reducing common mode noise especially from power line interference [3], [5].

Even with these improvements there are still some problems. Most designs use fixed transistor sizing and ideal input signals which do not represent real human body conditions. Because of this the simulation results may not match real world performance. Also improving gain without increasing power is still a challenge.

In this work we design a low power low noise CMOS ECG front end circuit. The main idea is to improve gain by optimizing transistor sizing. We also design a body modeling circuit to simulate real human body conditions. This helps in getting more realistic results. The complete system includes instrumentation amplifier OTA DRL circuit and filter stages. The results show that the proposed design gives good performance and can be used for wearable biomedical applications.

II. Methodology and Design

The proposed ECG front-end system is designed using different blocks which work together to amplify and clean the

input signal. The main parts of the system are the operational transconductance amplifier (OTA), instrumentation amplifier, DRL circuit, filter stages, and body modeling circuit.

First, the OTA is designed as the core amplification block of the analog front-end. It provides high gain and further amplification. The overall gain of the OTA depends on transconductance and output resistance as discussed in the next section.

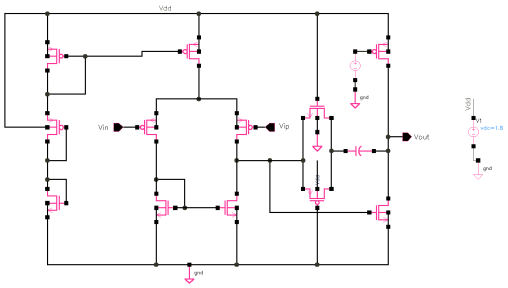


Fig. 1: Two stage OTA schematic

The designed OTA is then integrated into the instrumentation amplifier stage to amplify the weak differential ECG signal and reduce common-mode noise. The gain of the instrumentation amplifier using capacitive feedback can be written as

$$A_v = \frac{C_1}{C_f} \tag{1}$$

where C_1 is the input capacitor and C_f is the feedback capacitor [1].

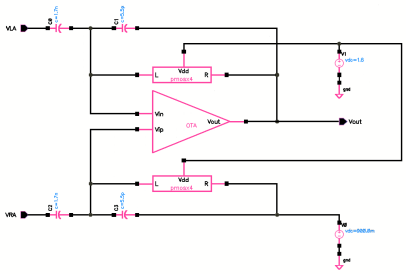


Fig. 2: Instrumentation amplifier block

The lower cutoff frequency of the amplifier depends on the pseudo resistor and feedback capacitor and is given by

$$f_L = \frac{1}{2\pi R_f C_f} \tag{2}$$

This helps in passing low frequency ECG signals and removing DC offset.

Next, a DRL circuit is used to reduce common-mode interference. It senses the common-mode voltage and feeds back an inverted signal. The reduction in common-mode voltage can be expressed as

$$V_{cm,out} = \frac{V_{cm,in}}{1 + A} \tag{3}$$

where A is the loop gain of the DRL circuit [5].

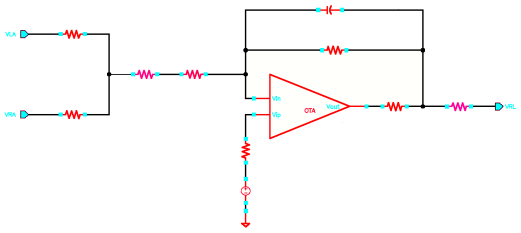


Fig. 3: Driven Right Leg circuit

Filter circuits are also used to remove unwanted noise. A high-pass filter is used to remove low-frequency noise, while a low-pass filter removes high-frequency noise. A notch filter is employed to suppress 50 Hz power-line interference.

One important part of this work is the body modeling circuit. Instead of using an ideal input signal, a body model consisting of resistance and capacitance elements representing skin and tissue impedance is used to emulate realistic human body conditions.

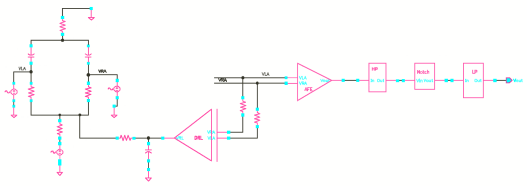


Fig. 4: Body modeling circuit

All the circuits are designed and simulated using Cadence Virtuoso and Spectre simulator. Different analyses like DC AC and transient simulations are performed to check gain bandwidth and noise performance of the system.

III. Results and Analysis

A. OTA Design and Performance

The OTA is the key gain stage of the proposed ECG front-end and is designed as a two-stage amplifier to achieve high voltage gain while maintaining low power consumption. The first stage employs a PMOS differential input pair (M7–M8), while the second stage is implemented as a common-source amplifier to provide additional gain. This architecture offers high overall gain together with sufficient output swing for low-amplitude ECG signals.

The transistor dimensions were optimized to improve both gain and noise performance. The input transistors (M7–M8) use a large device area to increase transconductance and reduce flicker noise, which is the dominant noise source at ECG frequencies. The active-load and gain-stage transistors (M0–M1 and M3) employ longer channel lengths to increase output resistance (Ro), thereby improving the intrinsic gain of each stage. Similarly, the PMOS current-source devices

(M10–M11) are implemented with longer channels to provide a high-impedance load and improve gain.

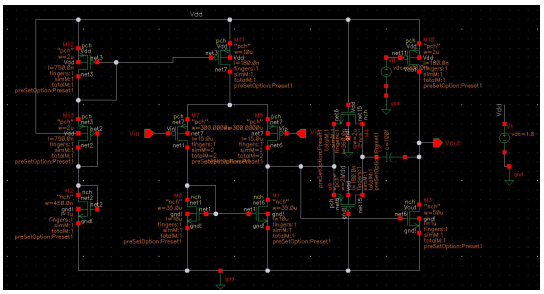


Fig. 5: Two stage OTA schematic

The bias network was also optimized through transistor sizing. The reference current mirror transistor (M12) uses a longer channel length than the remaining bias devices, reducing channel-length modulation and improving current-source accuracy. This results in a more stable bias current, leading to improved operating-point stability and reduced gain variation.

The overall voltage gain of the OTA can be written as

$$A_v = (g_{m1} r_{o1})(g_{m2} r_{o2}) \tag{4}$$

where g_m is the transconductance and r_o is the output resistance of each stage.

The transconductance of MOS transistor in saturation is given by

$$g_m = \sqrt{2\mu C_{ox} \frac{W}{L} I_D} \tag{5}$$

This shows that the transconductance depends on transistor size and drain current. In this work transistor sizing is optimized by increasing the width to length ratio. This improves g_m and increases the gain of the OTA.

For low power operation the transistors are biased in sub-threshold region. In this region the transconductance is given by

$$g_m = \frac{I_D}{nV_T} \tag{6}$$

where V_T is thermal voltage and n is a constant. This allows high gain even at low current which is useful for wearable devices.

Noise is also an important factor in ECG circuits. Since ECG signals are low frequency flicker noise becomes dominant. The flicker noise is given by

$$V_n^2 = \frac{K}{C_{ox} W L f} \tag{7}$$

From this equation we can see that increasing transistor area reduces noise. So larger input transistors are used to improve noise performance [3].

To make the OTA stable a compensation capacitor is used between the two stages. This introduces a dominant pole and improves phase margin. The unity gain bandwidth of the OTA can be written as

$$GBW = \frac{g_{m1}}{2\pi C_c} \tag{8}$$

where C_c is the compensation capacitor.

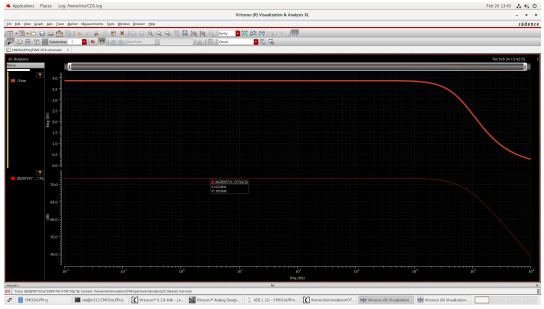


Fig. 6: Gain of OTA

From simulation results the OTA shows high gain and stable frequency response. The gain improvement is mainly due to optimized transistor sizing. This makes the OTA suitable for amplifying low amplitude ECG signals.

B. Instrumentation Amplifier

The instrumentation amplifier is the first stage of the ECG front end system. It is used to amplify the weak signal coming from the electrodes. Since the ECG signal is very small this stage is very important for proper signal acquisition.

The main advantage of instrumentation amplifier is that it provides high input impedance and high common mode rejection ratio. High input impedance helps in avoiding loading effect on the signal source. High common mode rejection helps in reducing noise which is present on both inputs such as power line interference [1].

In this design capacitive feedback is used to control the gain. The mid band gain of the instrumentation amplifier can be written as

$$A_v = \frac{C_1}{C_f} \tag{9}$$

where C_1 is the input capacitor and C_f is the feedback capacitor. This method helps in maintaining stable gain and also removes DC offset present at the input.

Pseudo resistors are used in the feedback path to provide very high resistance value. These resistors are implemented using MOS transistors and can have resistance in the order of tera ohms. This helps in achieving very low cutoff frequency.

The lower cutoff frequency of the amplifier is given by

$$f_L = \frac{1}{2\pi R_f C_f} \tag{10}$$

where R_f is the pseudo resistor. This ensures that low frequency ECG signals are passed without attenuation.

The input referred noise of the amplifier is also an important parameter. It can be expressed as

$$V_{n,in}^2 = \left(\frac{C_1 + C_f + C_{par}}{C_1} \right)^2 V_{n,ota}^2 \tag{11}$$

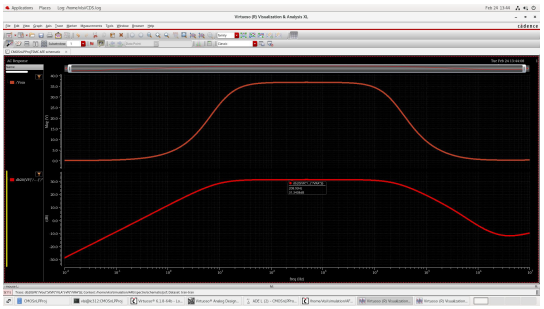


Fig. 7: Overall gain response of the analog front-end (AFE), demonstrating amplification of low-amplitude ECG signals within the desired bandwidth.

where C_{par} represents parasitic capacitance and $V_{n,ota}$ is the noise of the OTA [3].

From simulation results the instrumentation amplifier shows good gain and helps in reducing noise at the input stage. It also removes DC offset and prepares the signal for further amplification by the OTA.

Overall the instrumentation amplifier improves signal quality and plays a key role in the ECG front end system.

C. DRL Circuit

The Driven Right Leg (DRL) circuit is used to reduce common mode noise in the ECG system. In practical conditions the human body picks up interference from power lines and other sources. This creates common mode voltage at the input which affects the signal quality.

The DRL circuit works by sensing the common mode voltage from the input signals. This voltage is then inverted using an amplifier and fed back to the body through a third electrode. This feedback helps in canceling the noise present in the system [5].

The reduction of common mode voltage depends on the loop gain of the DRL circuit. The output common mode voltage can be written as

$$V_{cm,out} = \frac{V_{cm,in}}{1+A} \tag{12}$$

where A is the gain of the feedback loop. Higher loop gain gives better noise reduction.

The common mode rejection ratio (CMRR) is improved due to the DRL circuit and is given by

$$CMRR = 20\log\left(\frac{A_d}{A_{cm}}\right) \tag{13}$$

where A_d is differential gain and A_{cm} is common mode gain.

A protection resistor is also used in the DRL circuit to limit the current flowing into the body. This is important for safety. The current through the body can be written as

$$I = \frac{V_{DD}}{R} \tag{14}$$

where R is the protection resistance. This ensures that the current is within safe limits.

To maintain stability a capacitor is added in the feedback path. This helps in improving phase margin and prevents oscillations in the system.

From simulation results it is observed that the DRL circuit reduces common mode noise effectively. This improves the quality of the ECG signal and increases the overall performance of the system.

D. Filters and Noise Reduction

Filter circuits are used in the ECG front end system to remove unwanted noise from the signal. Since ECG signals are very weak they can be easily affected by different types of noise like baseline drift muscle noise and power line interference.

In this design three types of filters are used which are high pass filter low pass filter and notch filter.

The high pass filter is used to remove low frequency noise such as baseline drift. This noise is usually caused by movement of electrodes or body motion. The cutoff frequency of the high pass filter is given by

$$f_c = \frac{1}{2\pi RC} \tag{15}$$

This filter allows frequencies above the cutoff value to pass and blocks very low frequency components.

The low pass filter is used to remove high frequency noise such as muscle noise and external interference. It allows low frequency ECG signals to pass and blocks higher frequency components. The cutoff frequency is also given by

$$f_c = \frac{1}{2\pi RC} \tag{16}$$

The notch filter is used to remove power line interference which is usually at 50 Hz. This type of noise is very common in ECG systems. The notch filter blocks a narrow range of frequencies around 50 Hz and allows other frequencies to pass.

The transfer function of a notch filter can be written as

$$H(s) = \frac{s^2+\omega_0^2}{s^2+\frac{\omega_0}{Q}s+\omega_0^2} \tag{17}$$

where ω_0 is the notch frequency and Q is the quality factor.



Fig. 8: Notch filter response at 50 Hz

By combining all these filters the ECG signal bandwidth is limited to the required range which is approximately 0.5 Hz

to 100 Hz. This helps in removing most of the unwanted noise and improves the quality of the signal.

From simulation results it is observed that the filters effectively remove noise and allow only useful ECG signal components to pass.

E. Overall System Performance

The proposed ECG front end system is simulated using Cadence Virtuoso and Spectre simulator. Different analyses such as DC AC and transient simulations are performed to evaluate the performance of the circuit.

The differential gain of the system is obtained from AC analysis. The gain is found to be around 79.54 dB which shows that the circuit is able to amplify very weak ECG signals effectively.

$$A_d = 79.54\,dB \tag{18}$$

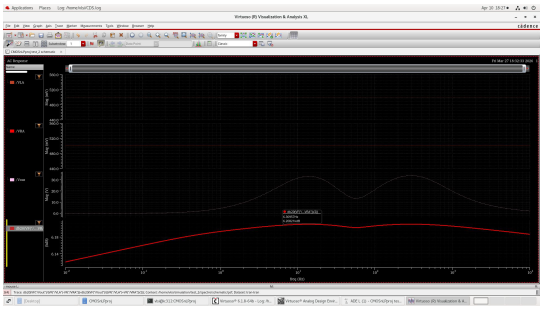


Fig. 9: Gain response of the complete test circuit including all stages, validating system-level performance of the proposed design.

The common mode gain is also measured to evaluate noise rejection capability. It is found to be around -34 dB. Using this value the common mode rejection ratio is calculated.

$$A_{cm} = -34\,dB \tag{19}$$

$$CMRR = A_d - A_{cm} \tag{20}$$

$$CMRR = 79.54 - (-34) = 113.54\,dB \tag{21}$$

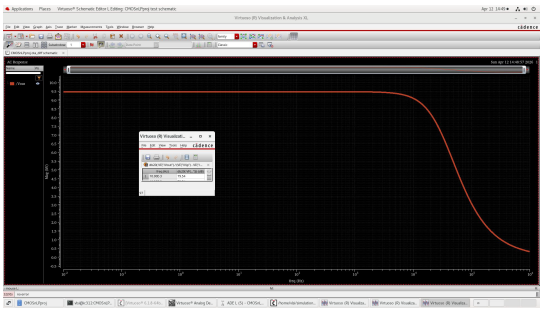


Fig. 10: CMRR calculation Normal AC plot

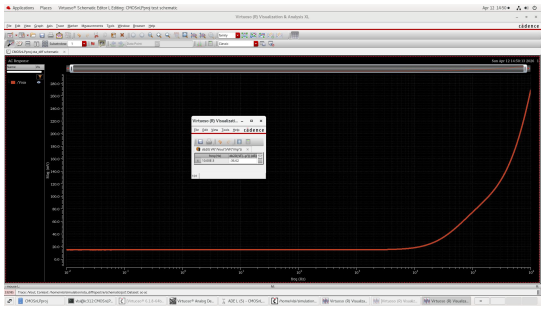


Fig. 11: CMRR calculation Common Mode plot

The bandwidth of the system is obtained from the frequency response. The lower cutoff frequency is around 198.4 mHz and the upper cutoff frequency is around 34 kHz.

$$BW = 198.4\,mHz \text{ to } 34\,kHz \tag{22}$$

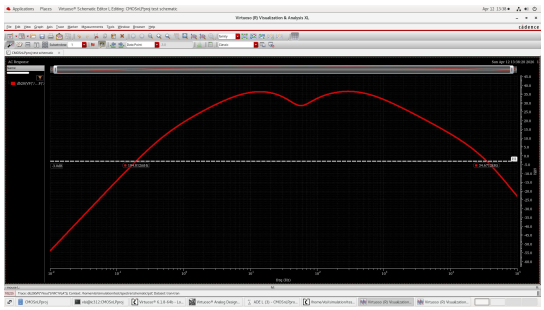


Fig. 12: Bandwidth calculation using -3 dB points

The unity gain frequency is also measured from the AC plot. It is found to be around 28.557 kHz. The gain bandwidth product is calculated using the gain and bandwidth values.

$$GBW = A_d \times BW \tag{23}$$

$$GBW \approx 9500 \times 34k \approx 323\,MHz \tag{24}$$

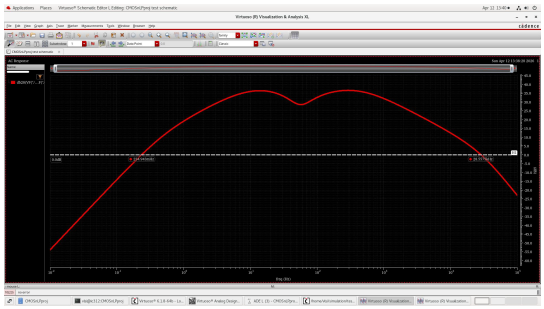


Fig. 13: Unity gain frequency plot

The transient analysis is used to calculate the slew rate of the circuit. From the waveform the voltage changes from 0.9 V to 1.0 V in a small time interval.

$$SR = \frac{\Delta V}{\Delta t} \tag{25}$$

$$SR = \frac{1.0 - 0.9}{5.8 - 5.11} \tag{26}$$

$$SR = 0.145\, V/ns \tag{27}$$

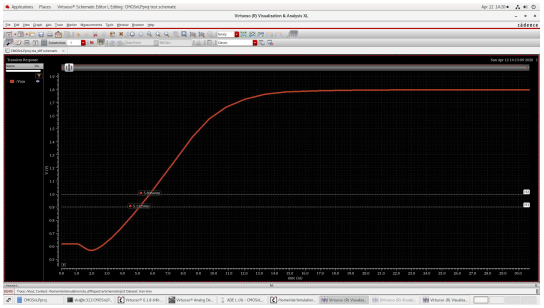


Fig. 14: Slew rate calculation from transient response

From all the results it is observed that the proposed design provides high gain good noise rejection and stable frequency response. The improvement in gain is mainly due to optimized transistor sizing. The circuit performance shows that it is suitable for ECG signal acquisition in wearable devices.

The results obtained from simulation show that the proposed ECG front end system performs well in terms of gain noise rejection and stability. The differential gain is high which means the circuit can amplify very weak ECG signals properly. This is important because ECG signals are very small in amplitude.

The CMRR value is also high which shows that the circuit is able to reject common mode noise effectively. This is mainly due to the use of instrumentation amplifier and DRL circuit. The DRL circuit plays an important role in reducing power line interference and improving signal quality.

The bandwidth of the system is wide enough to cover the required ECG signal range. Since ECG signals are mainly in the range of 0.5 Hz to 100 Hz the designed system is able to pass all useful signal components without distortion.

One of the main contributions of this work is transistor sizing optimization in the OTA. By properly choosing the width and length of transistors the gain is improved and noise is reduced. This shows that device level optimization can have a strong impact on overall circuit performance.

Another important contribution is the body modeling circuit. In many designs ideal input signals are used which do not represent real conditions. In this work the body model helps in testing the system under more realistic conditions. This gives better confidence in the simulation results.

The slew rate obtained from transient analysis shows that the circuit responds properly to input changes. This indicates good dynamic performance of the system.

However there are still some limitations. The power consumption can be further reduced for better performance in wearable devices. Also noise performance can be improved by using more advanced techniques.

Overall the proposed design shows good performance and meets the requirements for ECG signal acquisition. The combination of circuit design and optimization techniques makes it suitable for practical biomedical applications.

In this work a low power low noise CMOS ECG front end system is designed and simulated. The system includes instrumentation amplifier OTA DRL circuit and filter stages to process weak ECG signals.

The results show that the proposed design provides high gain good noise rejection and stable frequency response. The differential gain and CMRR values indicate that the circuit can amplify small signals and reduce common mode noise effectively.

The main contribution of this work is the optimization of transistor sizing in the OTA. This helps in improving gain and reducing noise without increasing power significantly. Another important contribution is the body modeling circuit which is used to simulate realistic human body conditions. This makes the simulation results more practical and reliable.

The overall performance of the system shows that it is suitable for ECG signal acquisition in wearable and portable biomedical devices. The design meets the basic requirements of low power operation and good signal quality.

In future work further improvements can be made to reduce power consumption and improve noise performance. More advanced circuit techniques can also be used to enhance the overall efficiency of the system.

REFERENCES

[1] Y. Laababid, K. E. Khadiri, and A. Tahiri, "Design of a low-power low-noise ecg amplifier for smart wearable devices using 180nm cmos technology," *WSEAS Transactions on Power Systems*, 2022.

[2] R. Kumar, "Signal processing techniques for removing noise from ecg signals," *Journal of Biomedical Engineering and Research*, 2019.

[3] J. Zhang, Z. Hong, S. Quan, and Z. Ruizhi, "A low-noise low-power amplifier with current-reused ota for ecg recordings," *IEEE Transactions on Biomedical Circuits and Systems*, 2018.

[4] M. Steyaert, W. Sansen, and C. Zhongyuan, "A micropower low-noise monolithic instrumentation amplifier for medical purposes," *IEEE Journal of Solid-State Circuits*, 1987.

[5] B. B. Winter and J. G. Webster, "Driven right leg circuit design," *IEEE Transactions on Biomedical Engineering*, 1983.